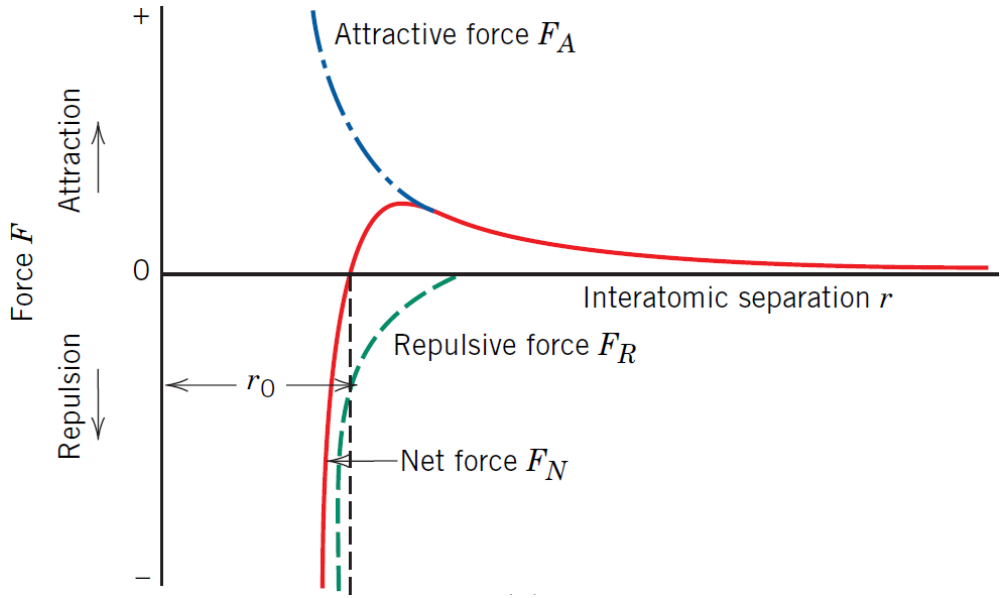
MT1011 Engineering Materials

Atomic bonding in solids

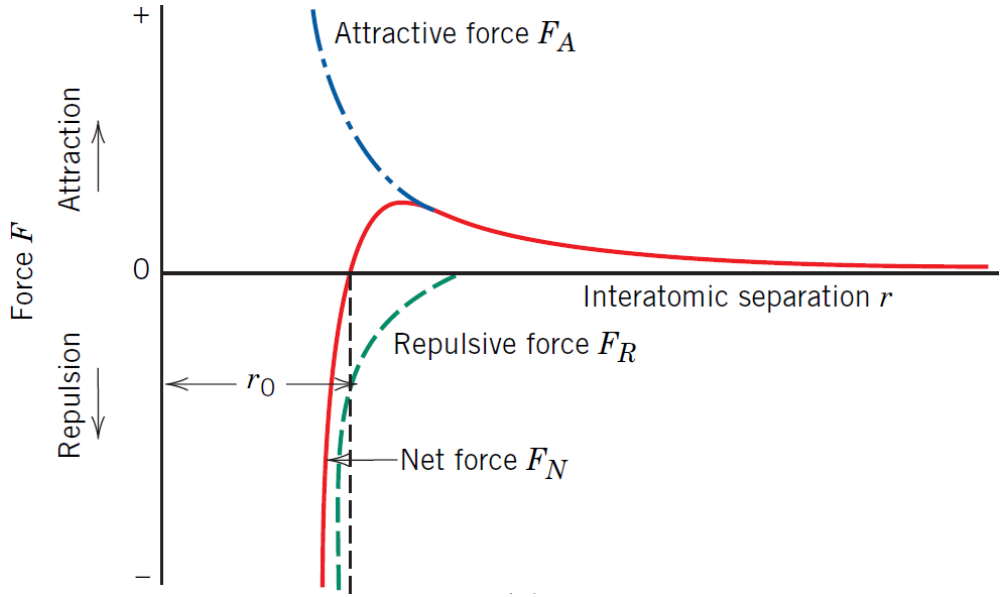
Dr Mudith Karunaratne

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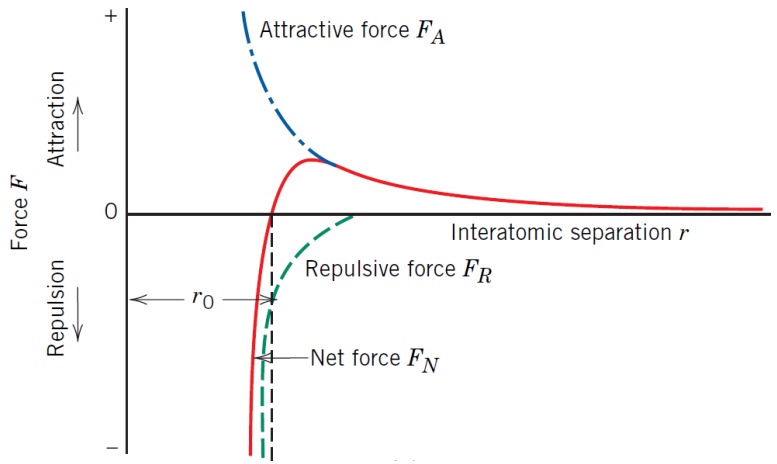
October 7, 2026



- Consider two isolated atoms interact as they are brought close together from a large separation.
- At large distances, interactions are negligible because the atoms are too far apart to have an influence on each other.
- At small separations, each atom exerts forces on the other.



- Two types of forces, attractive F_A and repulsive F_R forces.
- The magnitude of each depends on the separation or interatomic distance r.
- F_A depends on the type of bonding between the two atoms.
- F_R is due to the interactions between the negatively charged electron clouds of the two atoms.
- F_R is important only at small r as the outer electron shells of the atoms begin to overlap.



- The net force F_N between the two atoms is given by

$$F_N = F_A + F_R$$

- When F_A and F_R are equal in magnitude but opposite in sign, there is no net force (equilibrium state).

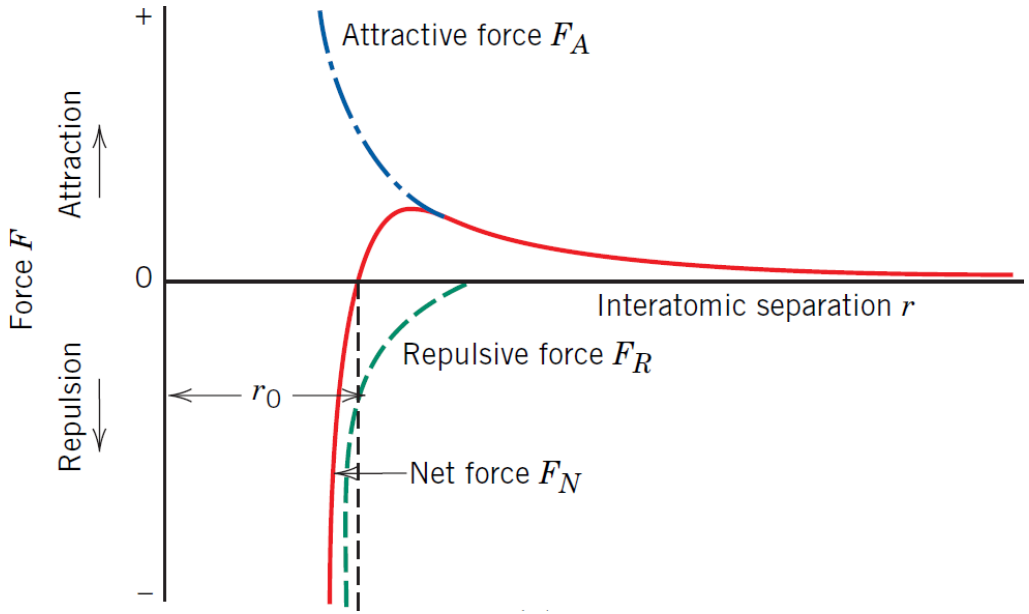
$$F_N = 0$$

- The centers of atoms are separated by the equilibrium spacing.
- For many atoms, the equilibrium spacing $r_0 = 0.3\,\text{nm}$.

Atomic bonding

Equilibrium position

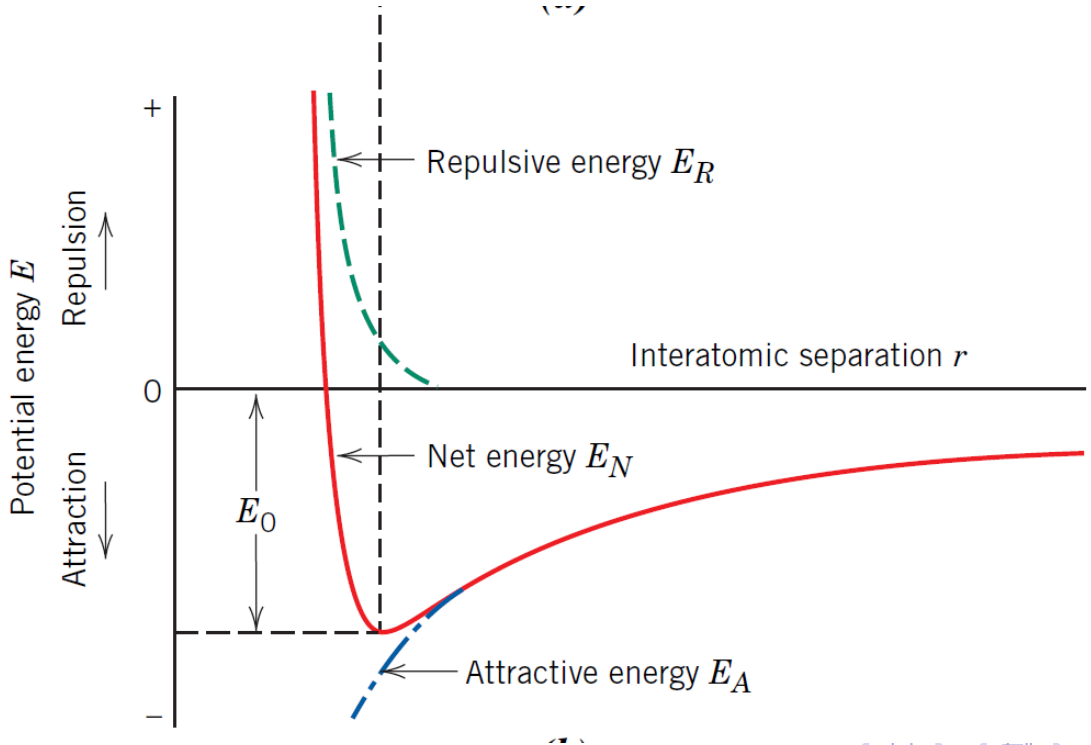
- Once the atoms are at equilibrium separation r_0, any attempt to move the two atoms farther apart is counteracted by the attractive force.
- Pushing the atoms closer together is resisted by the increasing repulsive force.
- The separation r_0 also relate to the minimum (potential) energy configuration.



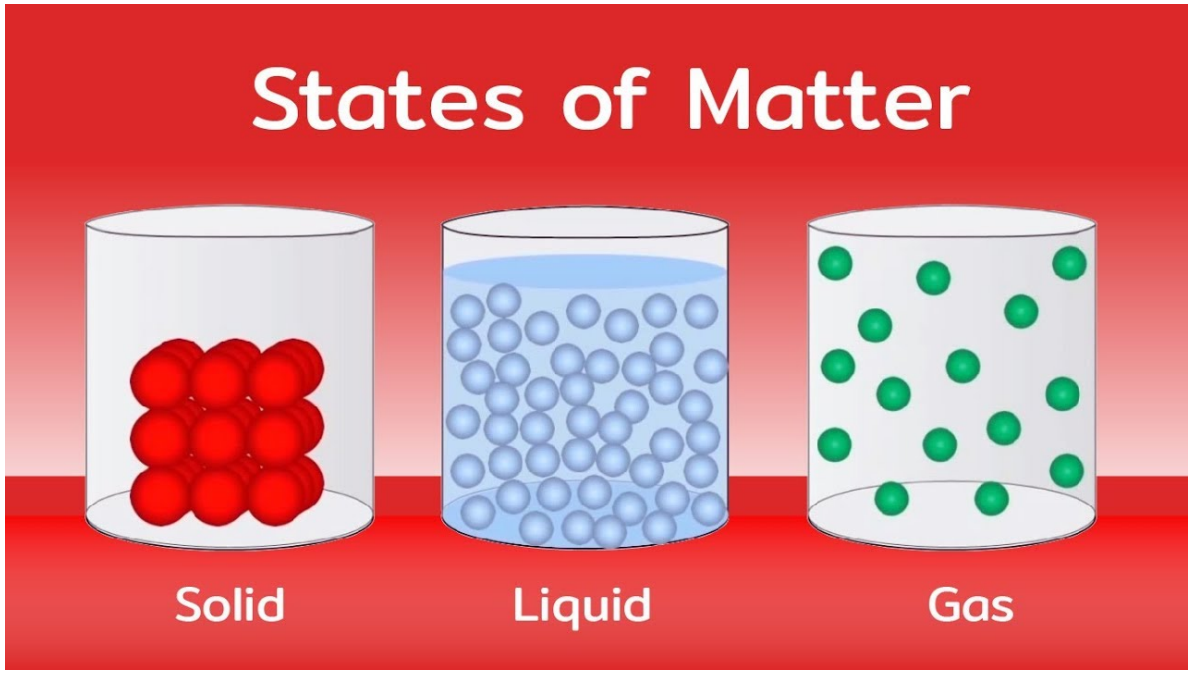
Atomic bonding

Bond energy

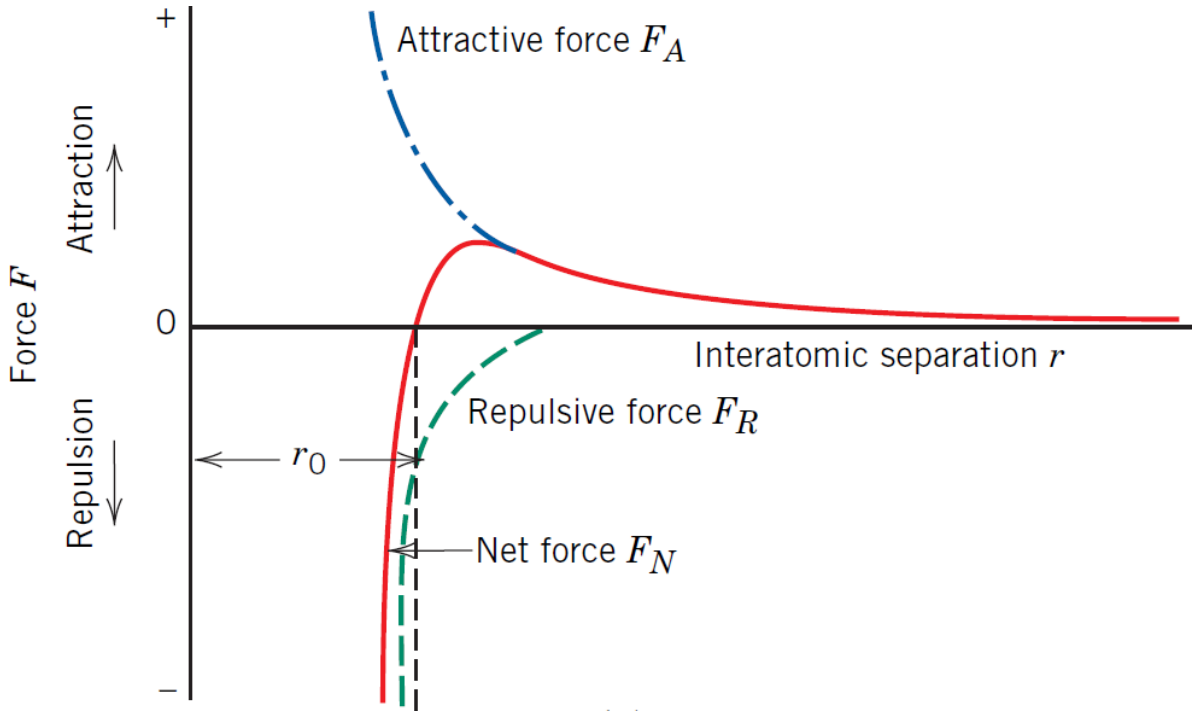
- Equilibrium spacing r_0 also correspond to the minimum energy configuration.
- $E_0 =$ **bond energy** for these two atoms at minimum point.
- Bond energy $=$ energy required for an infinite separation.
- Analysis can be extended to solids with more than two atoms.
- Bonding energy and energy-curve depend on the material.



- Many material properties depend on E_0, the curve shape, and type of the bonding.
- Materials with large bond energies have high melting points.
- Solids are formed for large bond energies.
- For small bond energies, the gaseous state is stable.
- At intermediate energies, liquid state is most stable.



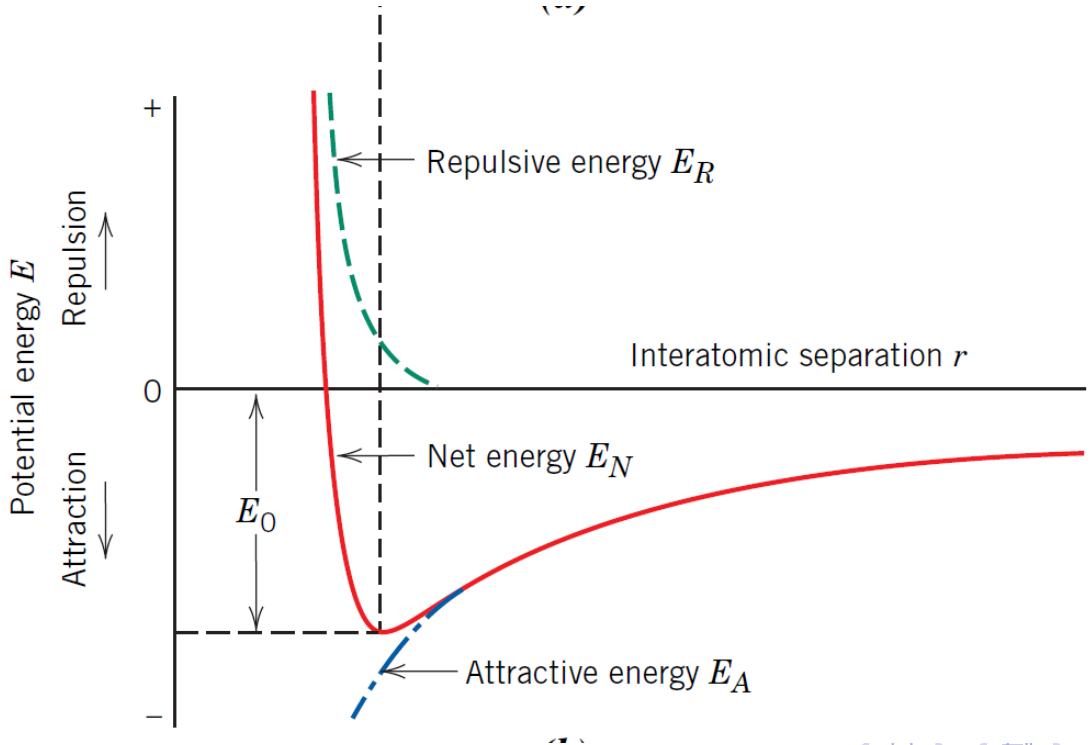
- Mechanical stiffness (or modulus of elasticity) is dependent on the shape of force-separation curve.
- The slope for a stiff material at $r = r_0$ position on the curve will be quite steep.
- Slopes are shallower for more flexible materials.



Atomic bonding

Bond energy and thermal expansion

- Thermal expansion of a material is related to the shape of the potential energy curve.
- A material with a deep and narrow energy curve, normally shows a low expansion coefficient.
- Such curves typically occur for materials with large bond energies.



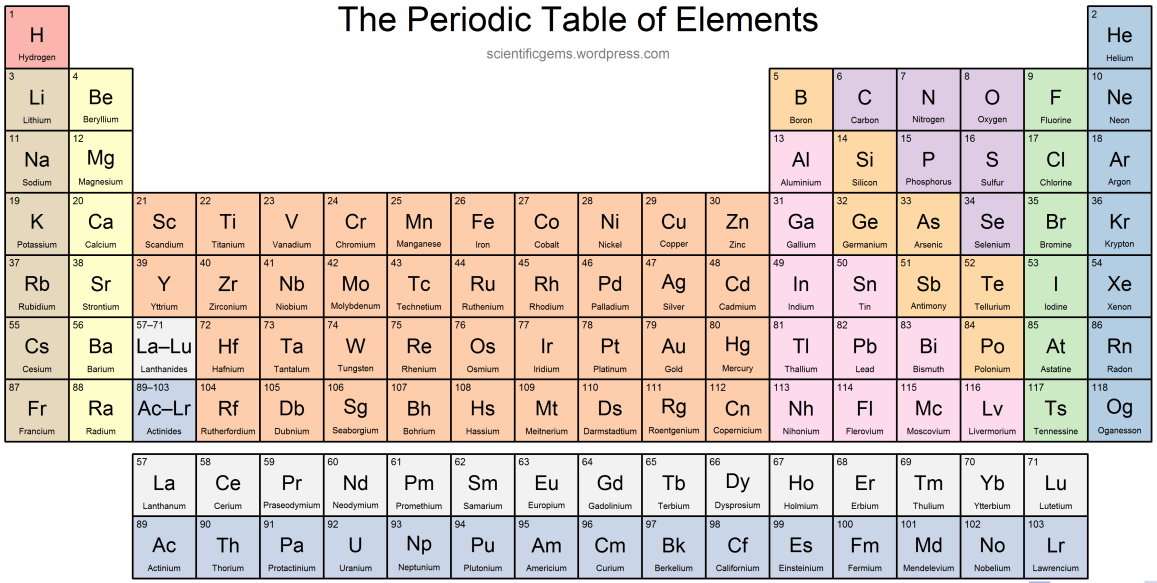
- Three different types of primary (or chemical) bond are found:
  1. Ionic
  2. Covalent
  3. Metallic
- Valence electrons are responsible for each type.
- Atoms achieve a stable electron structure (like those of the inert gases) by completely filling the outermost electron shell.
- Primary bonds arise due to this tendency.
- Type of the bond depends on the electron structure of atoms.



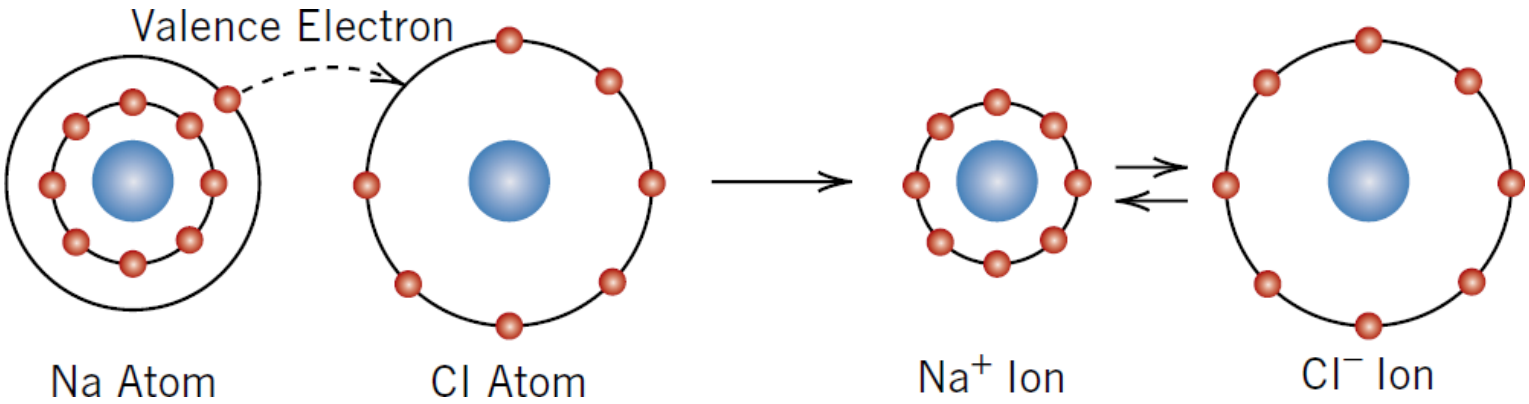
Atomic bonding

Ionic bonding

- Ionic bonds always found in compounds made between metallic and non-metallic elements (between elements at either end of the periodic table).
- Metallic atoms give up their valence electrons easily to the non-metallic atoms.
- Thus, all the atoms get stable configurations.
- The atoms also get an electrical charge and become ions.
- The attractive bonding forces are coulombic.



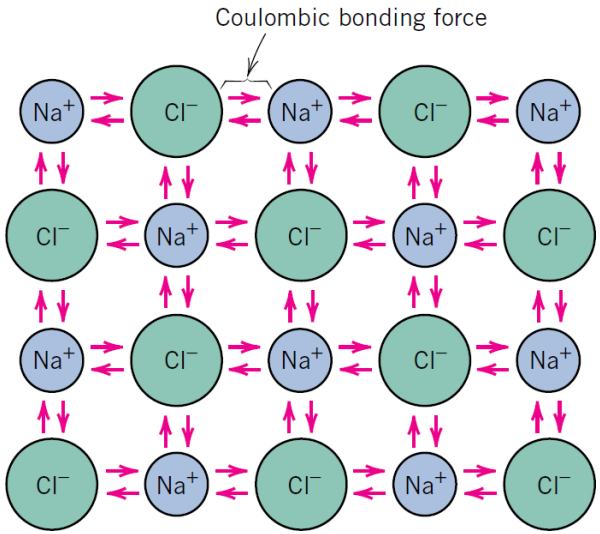
- In NaCl, a Na atom can assume the electron structure of neon.
- It will have a net single positive charge (with a reduction in size) by a transfer of one valence $3s$ electron to a Cl atom.
- After such a transfer, the Cl^- ion acquires a net negative charge and an electron configuration identical to that of argon.
- it is also larger than the Cl atom.



Atomic bonding

Ionic bonding

- Ionic bonding is nondirectional.
- Magnitude of the bond is equal in all directions around an ion.
- Hence, for stability, all positive ions must have negatively charged nearest neighbour ions in a 3-D scheme.
- Ionic bonding is found in ceramic materials, which are characteristically hard and brittle.
- These are electrically and thermally insulating.
- these properties are due to the electron configurations and/or the nature of the ionic bond.

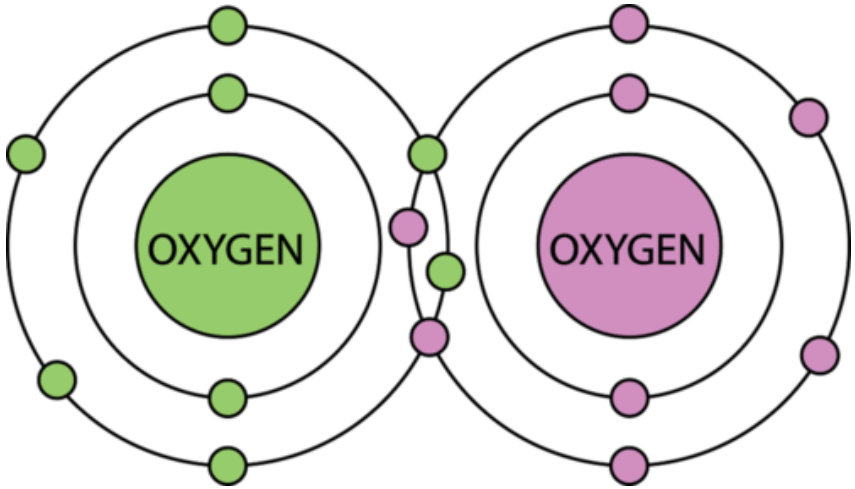


- Bonding energies for ionic solids, range between 600 and $1500\,\mathrm{kJ/mol}$.
- These energies are relatively large.
- Therefore these have high melting temperatures.

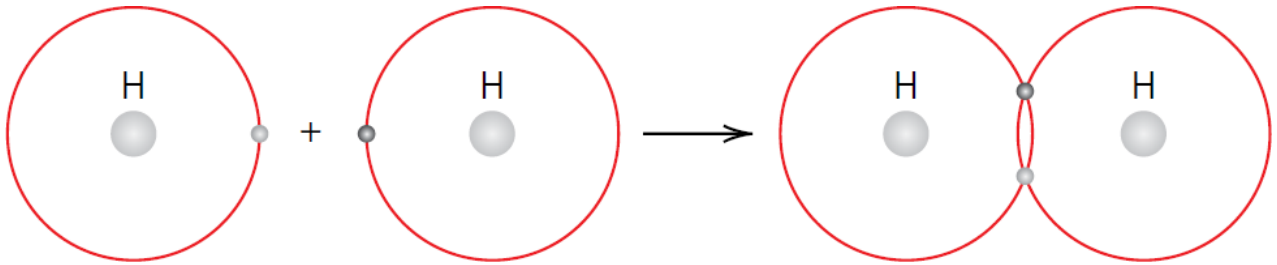
Substance	*Bonding Energy (kJ/mol)*	*Melting Temperature (°C)*
	Ionic	
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF_2	1548	1418



- Covalent bonding occurs between atoms with small differences in electronegativity.
- Usually, they occur near one another in the periodic table.
- Stable electron configurations are achieved by sharing electrons between neighbouring atoms.
- Two covalently bonded atoms will each contribute at least one electron to the bond.
- The electrons participating in the bond may be considered to belong to both atoms.



- The H atom has a single $1s$ electron.
- It can acquire a He electron configuration when two H atoms share each other's single electron.
- There is overlapping of electron orbitals in the region between the two bonding atoms.
- The covalent bond is *directional*.
- This is because the bonding is between specific atoms and may exist only in the direction between electron sharing atoms.



- Many nonmetallic elemental molecules (e.g., Cl_2, F_2), as well as molecules containing dissimilar atoms, such as CH_4, H_2O, HNO_3, and HF are covalently bonded.
- Covalent bonding is found in elemental solids such as diamond (C), Si, and Ge.
- Solid compounds having elements on the right side of the periodic table, are also covalently bonded, e.g. GaAs, InSb and SiC.
- Covalent bonds may be very strong, as in diamond.
- Diamond is very hard and has a high melting temperature, $3550\,^{\circ}$C.
- The bonds may also be very weak, as with Bi, which melts at about $270\,^{\circ}$C.

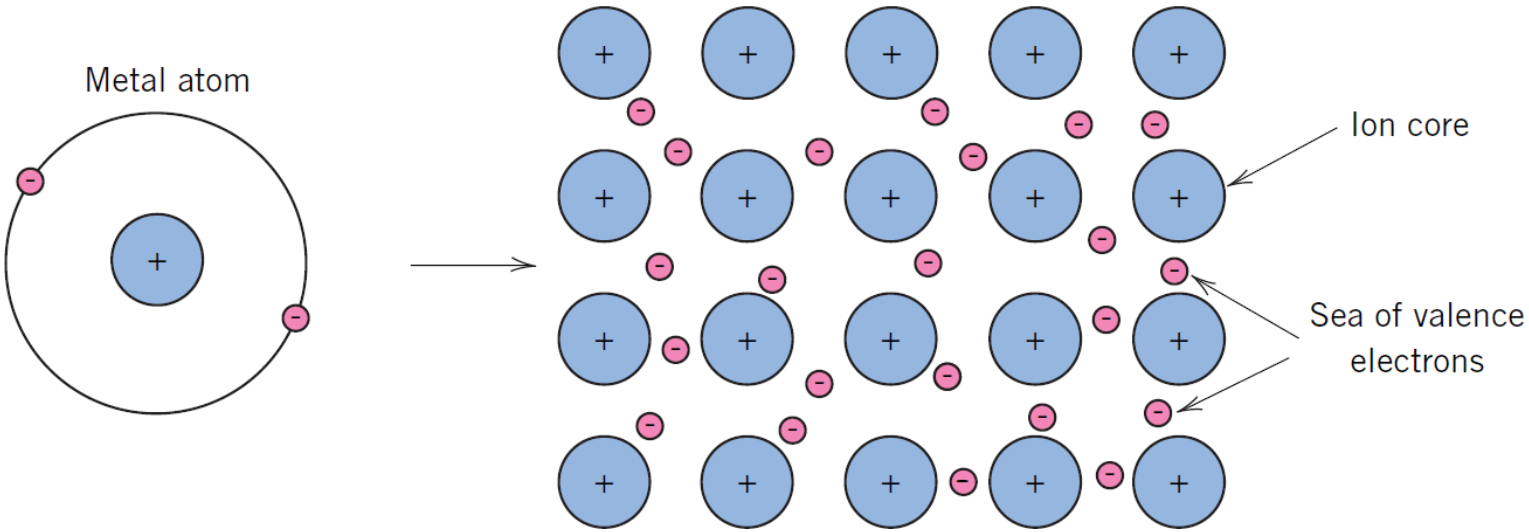


- Because electrons in covalent bonds are tightly bound to the bonding atoms, most covalent materials are electrical insulators.
- In some cases, they are semiconductors.
- Mechanical behaviours vary widely: strong, weak, brittle and ductile behaviour can be seen.
- It is difficult to predict the mechanical properties of on the basis of bonding characteristics.

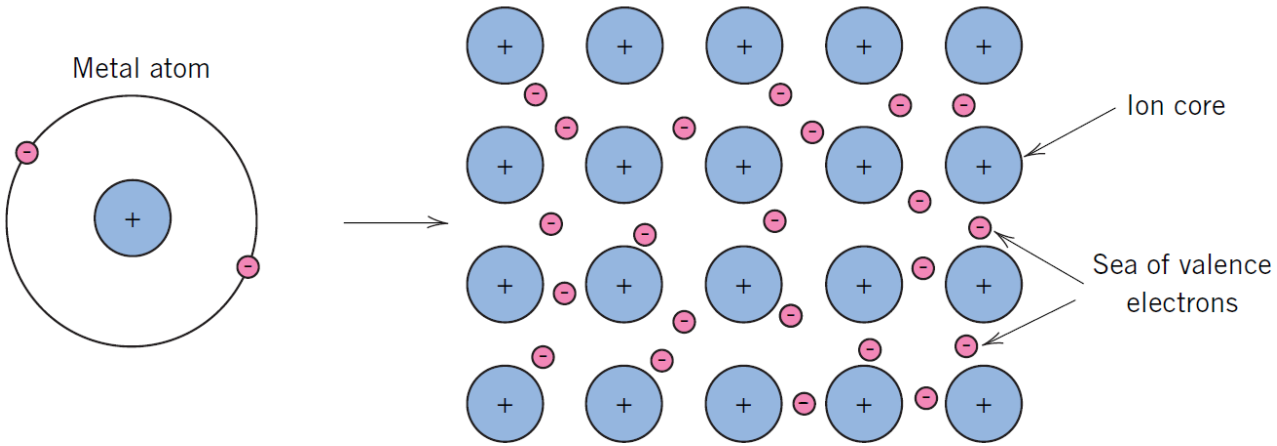
Substance	*Bonding Energy (kJ/mol)*	*Melting Temperature (°C)*
	Ionic	
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF_2	1548	1418
	Covalent	
Cl_2	121	−102
Si	450	1410
InSb	523	942
C (diamond)	713	>3550
SiC	1230	2830



- Metallic bonding is found in metals and alloys.
- According to a simple model, the valence electrons are not bound to any particular atom in the solid.
- They are free to drift throughout the entire metal.
- They belong to the metal as a whole, or forming a *sea of electrons* or an *electron cloud*.



- The remaining nonvalence electrons and atomic nuclei form ion cores.
- Mutually repulsive electrostatic forces exist between charged ion core atoms.
- The free electrons shield the positively charged ion cores from these forces.
- The metallic bond is nondirectional in character.
- The free electrons act as a *glue* to hold the ion cores together.



Substance	*Bonding Energy (kJ/mol)*	*Melting Temperature (°C)*
	Ionic	
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF_2	1548	1418
	Covalent	
Cl_2	121	−102
Si	450	1410
InSb	523	942
C (diamond)	713	>3550
SiC	1230	2830
	Metallic	
Hg	62	−39
Al	330	660
Ag	285	962
W	850	3414



- Bonding in metals may be weak or strong.
- Bond energies range from $62\,\text{kJ/mol}$ (Hg) to $850\,\text{kJ/mol}$ (W).
- Melting temperatures are $-39\,^{\circ}\text{C}$ (Hg) and $3414\,^{\circ}\text{C}$ (W).
- Metallic bonding is found in the periodic table for Group IA and IIA elements and for all elemental metals.
- Metals are good conductors of both electricity and heat because of their free electrons.
- At room temperature, most metals fail in a ductile manner.
- Fracture occurs after they undergo a significant amount of permanent deformation.
- Plastic deformation mechanism which is related to the metallic bond.



Atomic bonding

secondary bond (van der Waals bond)

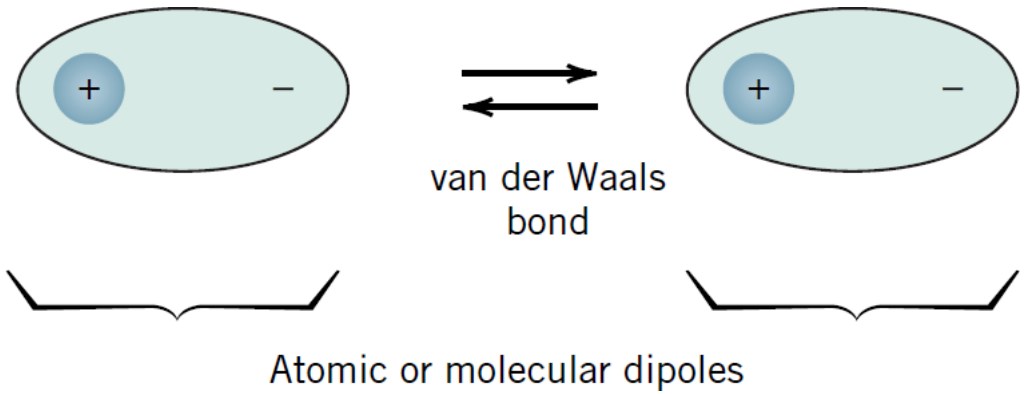
- The van der Waals bonds are weak compared to the primary bonds (compare 4 and $30\,\mathrm{kJ/mol}$).
- They exist between all atoms or molecules.
- Their presence may not be obvious if stronger primary bonds are present.
- They are more observable in inert gases with stable electron structures (when no primary bonding present).
- These are possible between atoms or groups of atoms, which themselves are joined together by primary (or intramolecular) ionic or covalent bonds.



Atomic bonding

secondary bond (van der Waals bond)

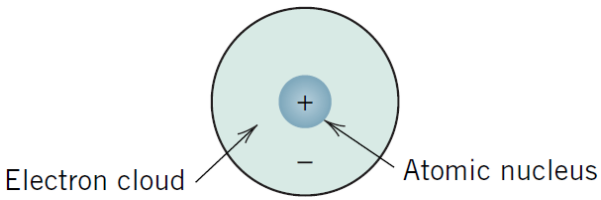
- Bonding is due to atomic or molecular *dipoles*.
- An electric dipole is created by separation of positive and negative parts of an atom or molecule.
- The bonding results from the coulombic attraction between the positive and negative ends of nearby dipoles.
- Dipole attractions occur between
  1. Induced dipoles
  2. Induced dipoles and polar molecules
  3. Polar molecules

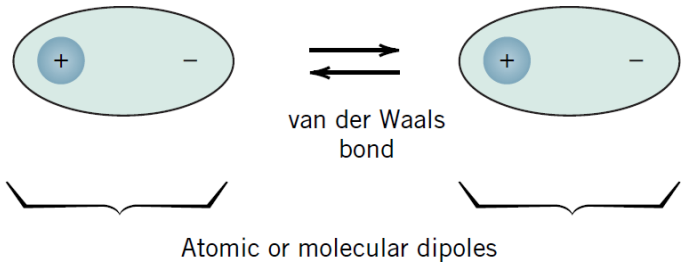


Atomic bonding

Fluctuation induced dipole bonding)

- A dipole may be induced in an atom (or molecule) which is normally electrically symmetric.
- This is because all atoms show vibrational motion that upset this electrical symmetry temporarily.
- A dipole can induce another dipole in a nearby molecule/atom.
- These dipoles are then weakly attracted or bonded.
- The attractive forces are temporary and fluctuate with time.
- However, they may occur between large numbers of atoms or molecules at any moment.

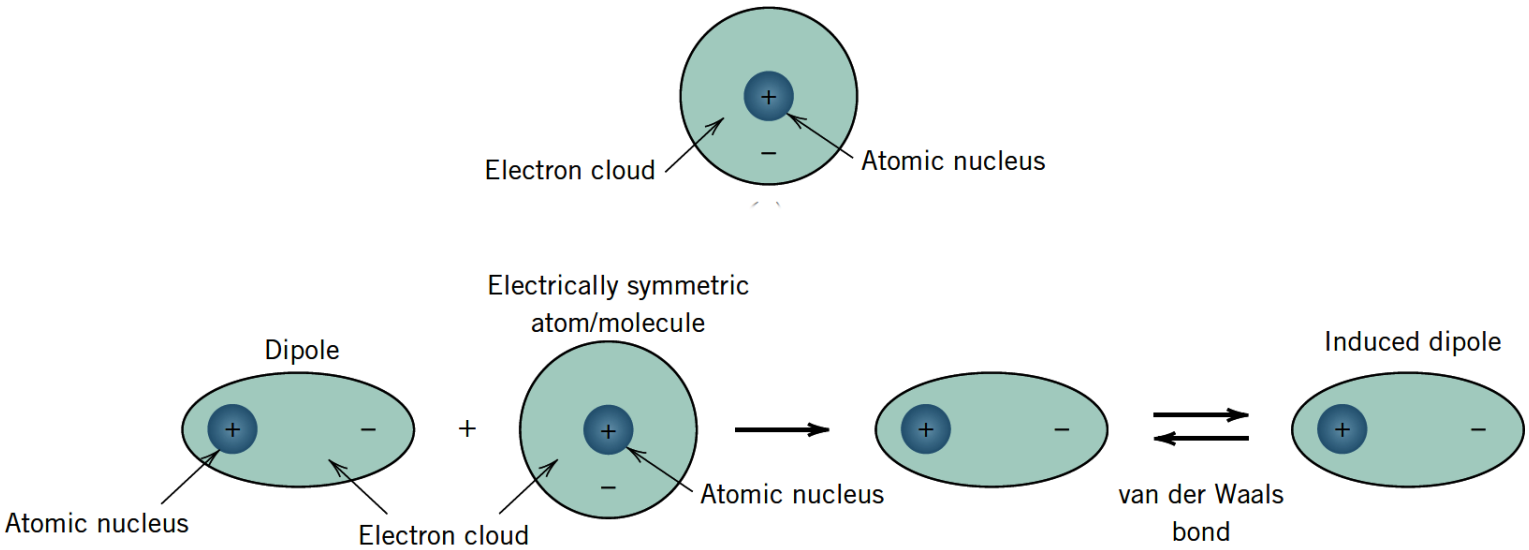




Atomic bonding

Fluctuation induced dipole bonding)

- The attractive forces are temporary and fluctuate with time.
- They may occur between large numbers of atoms or molecules at any moment.



Atomic bonding

Fluctuation induced dipole bonding)

- The liquefaction (some times even solidification) of the inert gases are possible due to this type of bonding.
- Electrically neutral and symmetric molecules such as H_2 and Cl_2 also behave this way.
- Of all possible intermolecular bonds, van der Waal bonds are the weakest.

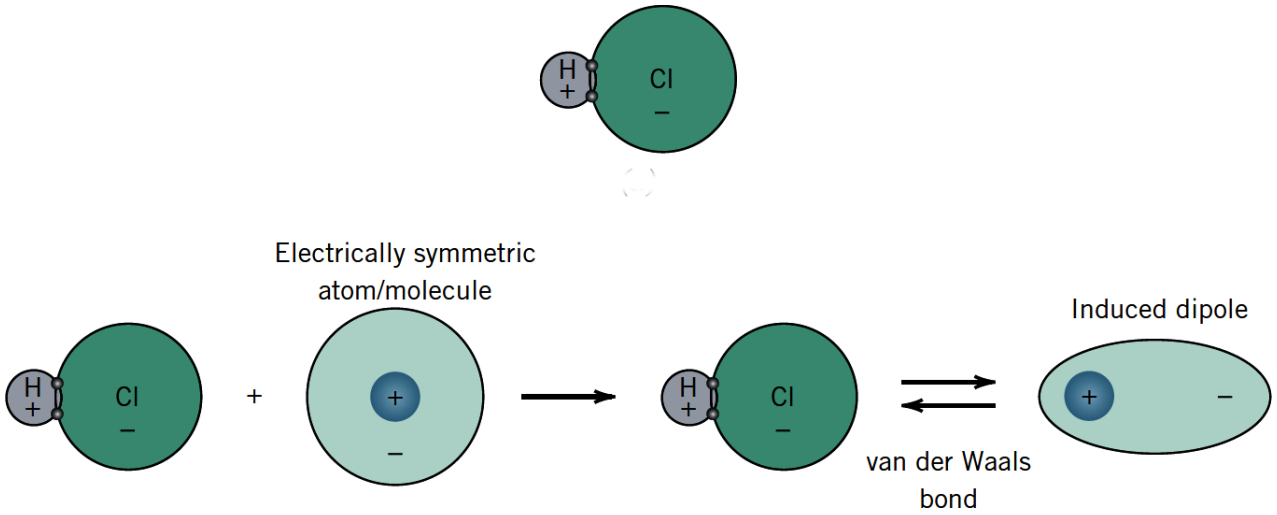
Substance	*Bonding Energy (kJ/mol)*	*Melting Temperature (°C)*
	Ionic	
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF_2	1548	1418
	Covalent	
Cl_2	121	−102
Si	450	1410
InSb	523	942
C (diamond)	713	>3550
SiC	1230	2830
	Metallic	
Hg	62	−39
Al	330	660
Ag	285	962
W	850	3414
	van der Waals[a]	
Ar	7.7	−189 (@ 69 kPa)
Kr	11.7	−158 (@ 73.2 kPa)
CH_4	18	−182
Cl_2	31	−101
	Hydrogen[a]	
HF	29	−83
NH_3	35	−78
H_2O	51	0



Atomic bonding

Polar molecule induced dipole bonding)

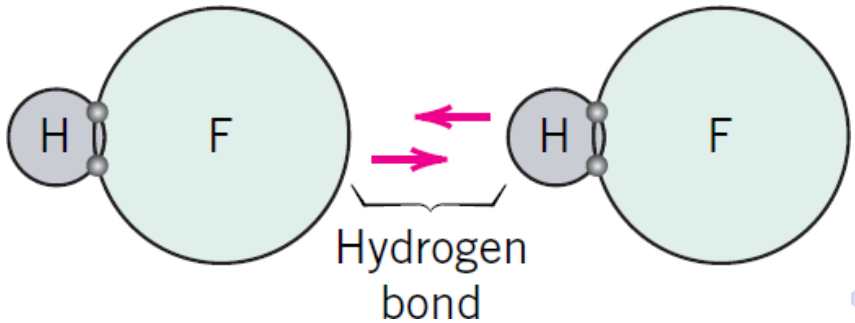
- Polar molecules have permanent dipoles due to non-symmetrical charge distributions.
- They can induce dipoles in nearby non-polar molecules and form bonds.
- Polar-molecule-induced dipole bonding is stronger than fluctuation induced dipole bonding.



Atomic bonding

Permanent dipole bonding

- In this type, bonding energies are significantly greater.
- The strongest secondary type is the *hydrogen bond* (special case of polar molecular bonding).
- It occurs between molecules where H is covalently bonded to F (e.g. HF), O (e,g, H_2O), or N (e.g. NH_3).
- The single electron of H is shared with the other atom.
- Thus, the H end of the bond is positively charged proton without any electrons.
- This end is capable of attracting a negative end of a nearby molecule.
- Melting and boiling temperatures for HF, NH_3, and H_2O are relatively high as a result of hydrogen bonding.



- For many real materials, the atomic bonds are mixtures of two or more of bond types.
- For mixed covalent–ionic bonds, an ionic character can be found in most covalent bonds and some covalent character in ionic bonds.
- Percent ionic character (%IC) of a bond between elements A and B (A being the most electronegative) may be given by:

$$\%\text{IC} = \left[1 - \exp\left\{-0.25(X_\text{A} - X_\text{B})^2\right\}\right]$$



Atomic bonding

Mixed bonding

